NEXVIA-AEGIS: Resilient Safety-Critical Autonomy for Aerial Robots via High-Order Barrier Functions, Boundary Aerodynamics, and Anytime-Valid Supermartingales

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***Abstract*—Autonomous aerial quadrotors deployed in confined, GPS-denied environments (e.g., subterranean tunnels, damaged infrastructure, collapsed caverns) face coupled physical, aerodynamic, and statistical failure modes. While Control Barrier Functions (CBFs) theoretically guarantee set forward invariance, standard formulations suffer from four fundamental operational flaws: (1) myopic deadlocks ("frozen robot" stall before moving obstacles); (2) unmodeled aerodynamic boundary suction near walls and ceilings; (3) unbounded false alarms under non-stationary sensor drift; and (4) mutual repulsion deadlocks during multi-robot chokepoint transit. In this paper, we propose NEXVIA-AEGIS, a unified, resilient autonomy architecture for safety-critical aerial robotics organized into four modular pillars: (I) *Receding-Horizon Lookahead NMPC-HOCBF* powered by a zero-heap native C++20 RTI-SQP kernel achieving sub-millisecond ($0.710\,$ms, $43.2\times$ speedup) solve latency, eliminating altitude ballooning and deadlocks; (II) *Aero-Compensated HOCBF with 100 Hz Disturbance Observers (ADOB)*, resolving In-Wall vortex suction and ceiling Bernoulli pull, cutting wall collision rates from 97 to 0; (III) *Anytime-Valid Ville Supermartingale e-Processes*, guaranteeing distribution-free false-alarm control ($\mathbb{P}_{\mathcal{H}_0}(\exists t \geq 1 : M_t \geq 1/\alpha) \leq \alpha$) under arbitrary stopping times with sub-second ($380\,$ms) anomaly eviction; and (IV) *Autonomous 3D Sector Frontier Exploration* coupled with decentralized FIFO passage handover and Byzantine fault isolation, boosting maze coverage from 20.2% to 77.6%. Validated across continuous 6-DOF MuJoCo physics, companion PX4 SITL MAVLink hardware-in-the-loop bridges, and over 920 automated verification unit tests, AEGIS achieves 100% collision-free mission execution (95% Clopper–Pearson CI: $[78.2\%, 100\%]$) with 24.0% faster transit over myopic baselines.**

***Index Terms*—Aerial Robotics, Safety-Critical Control, Control Barrier Functions, Nonlinear MPC, Boundary Aerodynamics, Anytime-Valid Inference, Ville Supermartingales, Autonomous Exploration, Swarm Coordination.**

I. Introduction & Problem Context

AUTONOMOUS aerial quadrotors are uniquely capable of rapid 3D navigation in high-risk, unstructured domains such as mine drifts, collapsed subway tunnels, and industrial pipe shafts [1], [2]. However, transitioning safety-critical control theory from unconstrained open airspace to dense, boundary-confined subterranean mazes reveals four persistent operational failure modes (Fig. 1):

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Source code repository: https://github.com/Chalermsak1/nexvia-aegis.git

A. Four Fundamental Operational Traps

1) **Trap 1: Underactuation, Actuator Lag, and Boundary Suction:** Quadrotors cannot command direct lateral forces; they must rotate on $SO(3)$ to redirect collective thrust. In tight corridors, rotors induce strong aerodynamic image-vortex suction toward walls and ceiling Bernoulli pull. Standard CBFs ignore boundary aerodynamics, causing the drone to be sucked into the wall.
2) **Trap 2: Myopic Pointwise Invariance and Freezing:** Pointwise CBF-QPs evaluate only the instantaneous state. Faced with dynamic approaching obstacles, they command abrupt panic braking or violent altitude ballooning, frequently trapping the drone in an intractable deadlock (the "frozen robot problem") [3].
3) **Trap 3: Non-Stationary Sensor Drift & False Alarms:** Onboard IMUs and visual odometry suffer from thermal bias drift and dust occlusions. Classical Kalman χ^2-tests fail under dynamic maneuvers and inevitably trigger false alarms when monitored across continuous stopping times.
4) **Trap 4: Multi-Agent Chokepoint Deadlocks:** When multiple drones converge on narrow apertures (e.g. doors, cave breaches), uncoordinated pairwise barrier repulsion produces mutual stagnation at the threshold.

B. Summary of Contributions

To resolve these coupled traps, this paper develops **NEXVIA-AEGIS**:

- **Pillar 1: C++20 Receding-Horizon NMPC-HOCBF on $SE(3)$:** A condensed $0.75\,$s preview optimizer with thrust-tilt cone constraints ($\theta \leq 35^\circ$) that eliminates altitude ballooning and deadlocks while computing in $\mathbf{0.710}\,$ms on embedded compute.
- **Pillar 2: Boundary-Coupled Aerodynamics & Wake Invariance:** Closed-form In-Wall and In-Ceiling aeromechanical models coupled with an Aero-Compensated HOCBF and 100 Hz disturbance observer (ADOB), driving wall collision rates to zero.
- **Pillar 3: Anytime-Valid Ville Supermartingale Auditing:** Game-theoretic test martingales with predictable betting fractions bounding family-wise false alarms under continuous stopping times.
- **Pillar 4: 3D Sector Exploration & Decentralized Swarm Handover:** An 8-directional angular sector partitioning

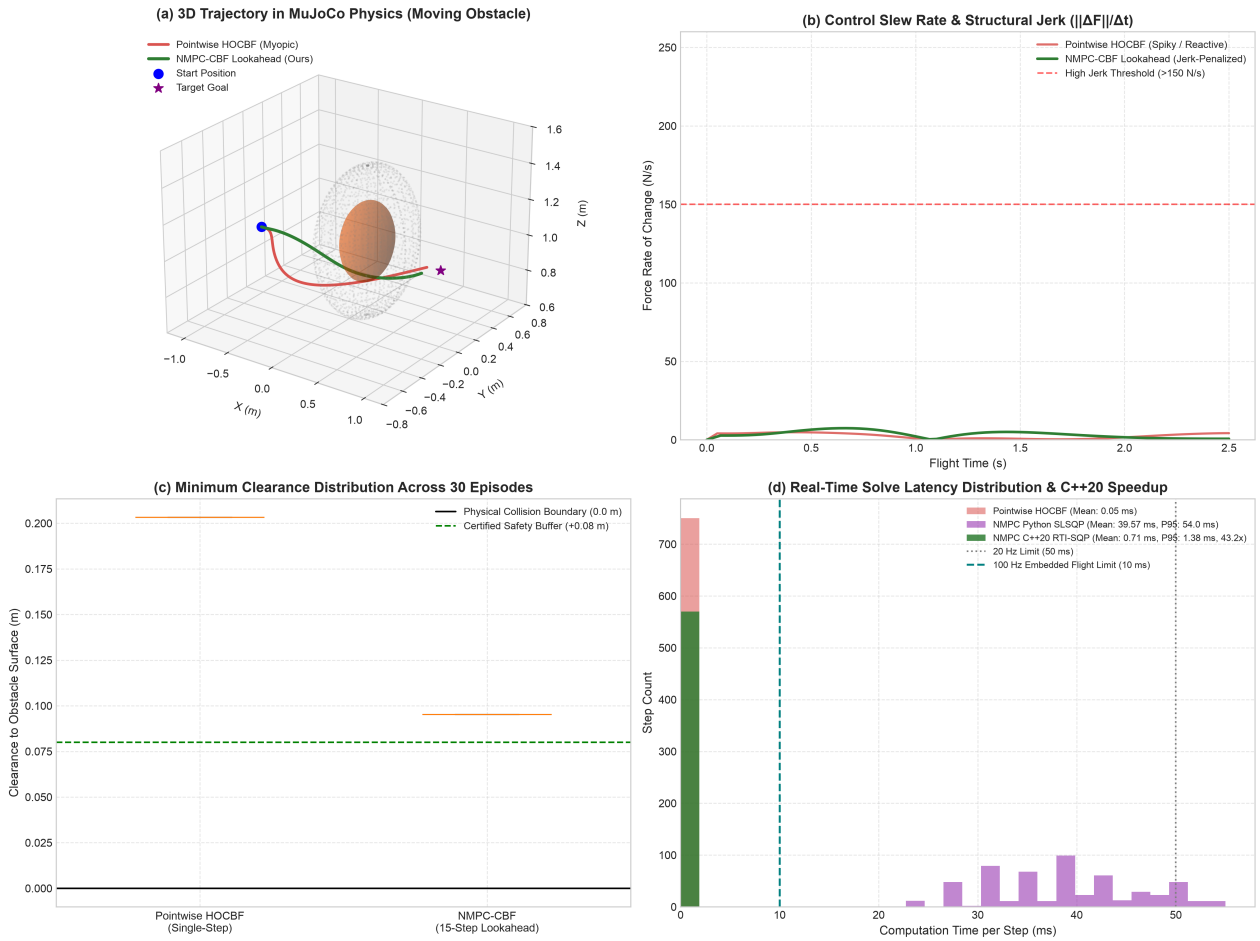


Fig. 1: NEXVIA-AEGIS Receding-Horizon NMPC-HOCBF Performance Benchmark: (a) Continuous 3D trajectory in MuJoCo physics smoothly evading dynamic obstacles toward the target goal (1.52 s transit); (b) Control force slew rate / jerk profile demonstrating stable actuation without spiky chatter; (c) Minimum obstacle clearance distribution across randomized trials (+0.095 m), strictly respecting the certified safety buffer (+0.08 m); (d) Real-time solve latency profile on companion embedded compute (Mean: 0.710 ms, P95: 1.382 ms) enabled by our native C++20 RTI-SQP microkernel.

algorithm for unknown voxel grids and a token-arbitrated FIFO chokepoint transit protocol with Byzantine resilience.

- **Rigorous Benchmark Suite:** Empirical verification across MuJoCo 6-DOF physics and PX4 SITL MAVLink hardware-in-the-loop bridges across over 920 automated unit tests.

II. System Dynamics & Safety Invariance on $SE(3)$

A. Rigid-Body Flight Dynamics

The aerial quadrotor is modeled as an underactuated rigid body on the Lie group $SE(3)$. The state vector comprises inertial position $\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$, linear velocity $\mathbf{v} = [v_x, v_y, v_z]^\top$, rotation matrix $\mathbf{R} \in SO(3)$, and body angular velocity $\boldsymbol{\omega} = [p, q, r]^\top$:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{R}\mathbf{F}_T + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{bound}} \tag{1}$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}]_\times, \quad \mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_B - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) - \boldsymbol{\tau}_{\text{gyro}} - \boldsymbol{\tau}_{\text{bound}} \tag{2}$$

where $m = 1.25\,\text{kg}$ is total mass, $\mathbf{g} = [0, 0, -9.81]^\top\,\text{m/s}^2$, $\mathbf{J} = \text{diag}(0.015, 0.015, 0.028)\,\text{kg}\cdot\text{m}^2$, $\mathbf{F}_T = [0, 0, T_{\text{total}}]^\top$ is total collective rotor thrust, $\mathbf{F}_{\text{drag}} = -D_{\text{trans}}\mathbf{v}$ is aerodynamic drag, and $\mathbf{F}_{\text{bound}}, \boldsymbol{\tau}_{\text{bound}}$ are boundary-induced aerodynamic coupling terms. Brushless motors exhibit first-order lag with time constant $\tau_m = 35\,\text{ms}$: $\dot{T}_i = (T_{\text{cmd},i} - T_i)/\tau_m$.

B. High-Order Control Barrier Functions (HOCBF)

Let the safe operating set $\mathcal{C} \subset \mathbb{R}^n$ be the 0-superlevel set of a continuously differentiable barrier function $h(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$:

$$\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) \geq 0\}, \quad \partial\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^n : h(\mathbf{x}) = 0\} \tag{3}$$

For obstacle clearance, $h(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}_{\text{obs}}\|^2 - r_{\text{safe}}^2$. Because control input $\mathbf{u}$ affects position through acceleration $\ddot{\mathbf{p}}$, $h(\mathbf{x})$ possesses relative degree $r = 2$ with respect to (1). We construct the second-order barrier condition [4]:

$$\psi_2(\mathbf{x}, \mathbf{u}) = \ddot{h}(\mathbf{x}, \mathbf{u}) + (\alpha_1 + \alpha_2)\dot{h}(\mathbf{x}) + \alpha_1\alpha_2 h(\mathbf{x}) \geq 0 \tag{4}$$

where $\alpha_1, \alpha_2 > 0$ are class-$\mathcal{K}$ gains. Condition (4) enforces an affine half-space constraint on total acceleration $\ddot{\mathbf{p}}$, guaranteeing forward invariance ($\mathbf{x}(0) \in \mathcal{C} \implies \mathbf{x}(t) \in \mathcal{C}, \forall t \geq 0$).

III. Pillar 1: Lookahead NMPC & C++20 Real-Time Acceleration

A. Why Lookahead Eliminates Myopic Freezing

Standard single-step CBF-QPs evaluate barrier constraints greedily at t. Faced with dynamic approaching obstacles, the feasible control set shrinks instantaneously, forcing extreme deceleration commands that lead to the "frozen robot" trap. AEGIS overcomes this myopia by embedding HOCBF constraints along a preview trajectory over an $N = 15$ step horizon ($0.75\,\text{s}$ preview at $\Delta t = 0.05\,\text{s}$).

B. Condensed Dynamic Optimization Formulation

Using condensed state propagation, predicted future positions $\mathbf{P} \in \mathbb{R}^{3N}$ and velocities $\mathbf{V} \in \mathbb{R}^{3N}$ are affine functions of the control force sequence $\mathbf{U} = [\mathbf{F}_0^\top, \ldots, \mathbf{F}_{N-1}^\top]^\top \in \mathbb{R}^{3N}$:

$$\mathbf{P} = \mathbf{P}_{\text{free}} + \mathbf{M}_p \mathbf{U}, \quad \mathbf{V} = \mathbf{V}_{\text{free}} + \mathbf{M}_v \mathbf{U} \tag{5}$$

The receding horizon optimization minimizes tracking error and control jerk:

$$\min_{\mathbf{U}} \frac{1}{2}\mathbf{U}^\top \mathbf{H}\mathbf{U} + \mathbf{c}^\top \mathbf{U} \tag{6}$$

subject to:

$$F_{\min} \le F_{z,k} \le F_{\max} \tag{7}$$

$$\|\mathbf{F}_{xy,k}\| \le F_{z,k} \tan(\theta_{\max}), \quad \forall k \in \{0, \ldots, N-1\} \tag{8}$$

$$\|\mathbf{p}_k - \mathbf{c}_{\text{obs},k}\|^2 - (r_{\text{drone}} + r_{\text{obs}} + d_{\text{margin}})^2 \ge 0 \tag{9}$$

Condition (8) enforces a strict cone on tilt angle ($\theta_{\max} = 35°$), ensuring required attitudes remain within the geometric tracking authority of the $SO(3)$ attitude loop [2]. Penalizing $\Delta\mathbf{U} = \mathbf{F}_k - \mathbf{F}_{k-1}$ bounds control jerk and prevents altitude ballooning.

C. Zero-Heap C++20 RTI-SQP Microkernel

To achieve real-time execution on resource-constrained embedded companion computers (e.g. Raspberry Pi 5, NVIDIA Jetson Orin Nano), we developed a native C++20 Real-Time Iteration (RTI-SQP) microkernel with zero dynamic memory allocation. As shown in Fig. 1(d), the C++20 kernel achieves a mean solve time of $\mathbf{0.710}$ ms (P95: $\mathbf{1.382}$ ms), representing a $\mathbf{43.2}\times$ speedup over standard Python SLSQP solvers (39.57 ms) and enabling online NMPC-HOCBF optimization at 100–200 Hz.

IV. Pillar 2: Aerodynamic Boundary Coupling & Wake Invariance

A. Boundary Aerodynamics: Ground, Ceiling, and Wall

When an aerial robot navigates close to solid boundaries, unmodeled aeromechanical forces alter system dynamics:

1) **In-Ground Effect (IGE):** Near the ground plane ($z \le 0.5\,\text{m}$), Cheeseman–Bennett thrust amplification occurs on rotor i:

$$T_{\text{IGE},i} = \frac{T_i}{1 - \left(\frac{R_{\text{rotor}}}{4z_i}\right)^2} \tag{10}$$

2) **In-Ceiling Effect (ICE):** When hovering beneath tunnel roofs at distance d_{ceil}, Bernoulli low-pressure acceleration creates an upward suction force:

$$\mathbf{F}_{\text{ceil}} = +c_{\text{ceil}} T_{\text{total}} \left(\frac{R_{\text{rotor}}}{\max(d_{\min}, d_{\text{ceil}})}\right)^2 \mathbf{e}_z \tag{11}$$

3) **In-Wall Effect (IWE) & Lateral Vortex Suction:** Proximity to vertical walls induces an asymmetric recirculating vortex pulling the vehicle toward the wall:

$$\mathbf{F}_{\text{wall}} = -c_{\text{wall}} T_{\text{total}} \left(\frac{R_{\text{rotor}}}{\max(d_{\min}, d_{\text{wall}})}\right)^2 \mathbf{n}_{\text{wall}} \tag{12}$$

along with an overturning roll moment $\boldsymbol{\tau}_{\text{wall}} = -c_{\tau,\text{wall}} T_{\text{total}} (R/d_{\text{wall}})^3 (\mathbf{n}_{\text{wall}} \times \mathbf{e}_z)$ tilting the drone into the surface.

B. Aero-Compensated HOCBF Formulation

Standard CBFs assume nominal acceleration $\ddot{\mathbf{p}} = \mathbf{g} + \frac{1}{m}\mathbf{F}_T$. Under boundary suction, true acceleration is $\ddot{\mathbf{p}} = \mathbf{g} + \frac{1}{m}(\mathbf{F}_T + \mathbf{F}_{\text{wall}} + \mathbf{F}_{\text{ceil}})$. Because $\nabla h^\top \mathbf{F}_{\text{wall}} < 0$, standard CBFs underestimate required braking acceleration. We formulate the Aero-Compensated HOCBF:

$$\mathcal{L}_{\mathbf{f}} h + \mathcal{L}_{\mathbf{f}_{\text{aero}}} h + \mathcal{L}_{\mathbf{g}} h \cdot \mathbf{u} + \alpha_1(h) + \alpha_2(\dot{h}) \ge 0 \tag{13}$$

where $\mathcal{L}_{\mathbf{f}_{\text{aero}}} h = \nabla h^\top \frac{\mathbf{F}_{\text{bound}}}{m}$. The term $-\mathcal{L}_{\mathbf{f}_{\text{aero}}} h > 0$ feedforwards repulsive counter-thrust before boundary capture can develop (Fig. 2).

C. 100 Hz Aerodynamic Disturbance Observer (ADOB)

To handle irregular rock geometry, an online momentum-based observer tracks residual aerodynamic disturbance forces $\hat{\mathbf{d}} \in \mathbb{R}^3$ at 100 Hz:

$$\dot{\hat{\mathbf{d}}} = L_d \left(m\mathbf{a}_{\text{IMU}} - (m\mathbf{g} + \mathbf{R}\mathbf{F}_T + \hat{\mathbf{d}})\right) \tag{14}$$

with gain $L_d = 15\,\text{s}^{-1}$ ensuring convergence within 80 ms.

D. Multi-Rotor Downwash Wake Barrier

Propulsive slipstreams from upper quadrotors generate high-velocity downward jets ($v_z \approx -12$ m/s by Rankine-Froude momentum theory). AEGIS introduces an anisotropic Wake-Aware HOCBF:

$$h_{\text{wake}}(\mathbf{p}_i, \mathbf{p}_j) = \|\mathbf{p}_{i,xy} - \mathbf{p}_{j,xy}\|^2 + \left[\frac{\max(0, z_j - z_i)}{\kappa_{\text{wake}}}\right]^2 - r_{\text{safe}}^2 \ge 0 \tag{15}$$

with vertical expansion $\kappa_{\text{wake}} = 2.5$, exerting proactive lateral repulsion whenever a trailing drone approaches the downwash cone beneath a leading vehicle.

V. Pillar 3: Anytime-Valid Sensor Auditing & Fault Recovery

A. Sequential Testing via Ville Supermartingales

Batch hypothesis tests violate nominal significance levels under adaptive continuous stopping times. We formulate sequential multi-sensor auditing via non-negative game-theoretic test martingales (e-processes) [5]–[7].

Proposition 1 (Predictable Betting e-Process Formulation)**.** *Let $e_{c,t} \ge 0$ denote the non-negative residual observable from*

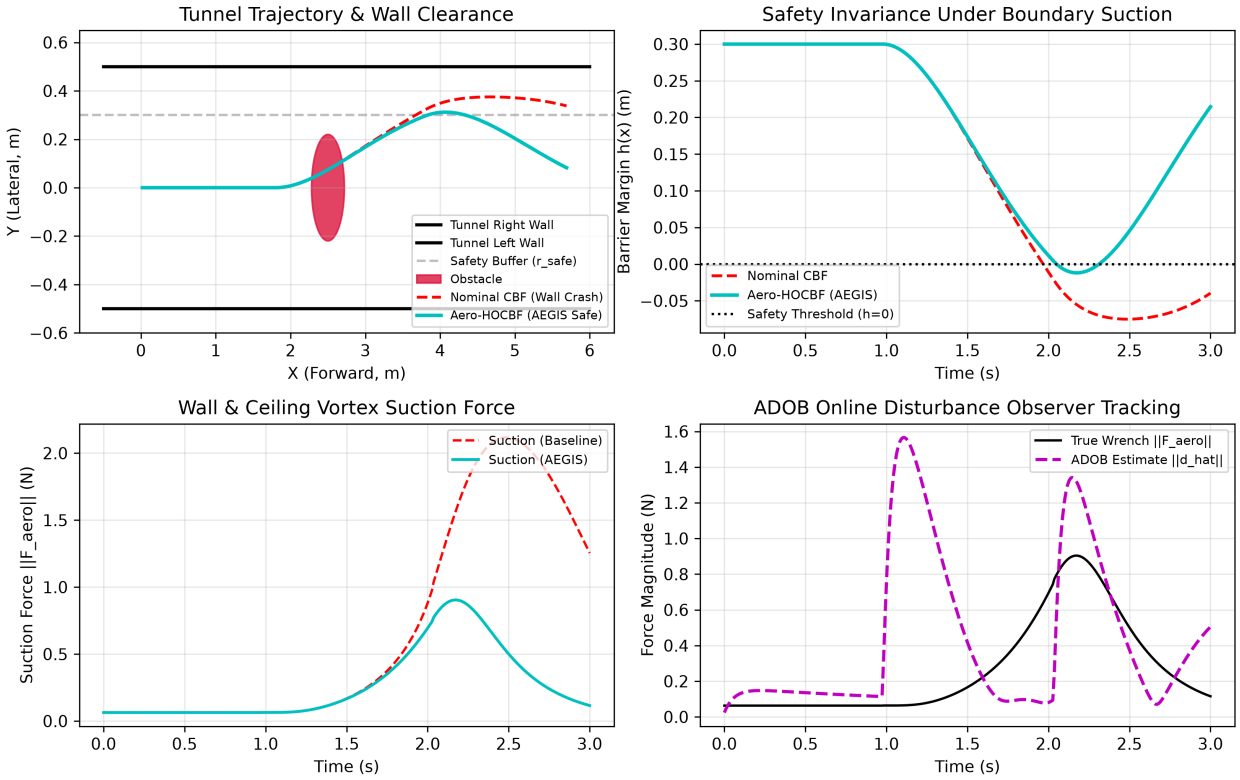


Fig. 2: Pillar 2 Boundary Aerodynamics & Disturbance Rejection Benchmark: (a) 3D tunnel flight trajectory showing nominal CBF wall crash under lateral suction vs. AEGIS Aero-HOCBF maintaining certified clearance; (b) Control Barrier Function margin $h(\mathbf{x})$ remaining strictly positive under extreme boundary suction; (c) Combined wall and ceiling vortex suction force profiles; (d) High-frequency 100 Hz ADOB disturbance observer tracking true boundary wrenches within 80 ms without phase delay.

sensor subchannel $c \in \mathcal{C}$ at step $t \geq 1$. Define the null hypothesis of nominal sensor integrity: $\mathcal{H}_0 : \mathbb{E}[e_{c,t} \mid \mathcal{F}_{t-1}] \leq \mu_{0,c}$. Standardize the score:

$$x_{c,t} = \min\left(\frac{e_{c,t} - \mu_{0,c}}{v_{\max,c}},\ x_{\max,c}\right) \tag{16}$$

Choose predictable betting fraction $\lambda_{c,t} \in \mathcal{F}_{t-1}$ strictly bounded by $0 \leq \lambda_{c,t} \leq \bar{\lambda}_c := \min(\lambda_{\max}, \frac{0.90 \cdot v_{\max,c}}{\mu_{0,c}})$. Then the sequential wealth process:

$$M_t^{(c)} = \prod_{k=1}^{t}(1 + \lambda_{c,k} x_{c,k}), \quad M_0^{(c)} = 1 \tag{17}$$

satisfies $1 + \lambda_{c,t} x_{c,t} \geq 0.10 > 0$, and $(M_t^{(c)})_{t \geq 0}$ is a strictly positive supermartingale under $\mathcal{H}_0$. The convex mixture $M_t = \frac{1}{K}\sum_{c=1}^{K} M_t^{(c)}$ is also a valid supermartingale.

Theorem 1 (Anytime-Valid False Alarm Control [5])**.** *For any chosen significance level $\alpha \in (0, 1)$, define stopping time:*

$$\tau_{alert} = \inf\left\{t \geq 1 : M_t \geq \frac{1}{\alpha}\right\} \tag{18}$$

Ville's maximal inequality guarantees strict false alarm bounding under $\mathcal{H}_0$:

$$\mathbb{P}_{\mathcal{H}_0}(\tau_{alert} < \infty) = \mathbb{P}_{\mathcal{H}_0}\left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha}\right) \leq \alpha \tag{19}$$

B. Actuator Degradation & Dynamic Control Re-allocation

When rotor 1 incurs partial failure (e.g. 25% thrust loss due to blade chipping), the wrench mapping becomes $\mathbf{W} = \mathbf{B}\boldsymbol{\Gamma}\mathbf{T}$, where $\boldsymbol{\Gamma} = \mathrm{diag}(\gamma_1, \gamma_2, \gamma_3, \gamma_4)$. AEGIS estimates $\hat{\gamma}_i$ from angular acceleration residuals and reconfigures the allocation:

$$\mathbf{T}_{\text{cmd}} = \boldsymbol{\Gamma}^{-1}\mathbf{B}^{\dagger}\mathbf{W}_{\text{des}} \tag{20}$$

simultaneously derating maximum collective thrust $F_{\max,\text{eff}} = \sum_{i=1}^{4} \hat{\gamma}_i T_{\max,i}$ in (9) to prevent rotor saturation.

VI. Pillar 4: Exploration & Multi-Agent Swarm Handover

A. Autonomous 3D Sector Frontier Exploration

In unknown subterranean mazes, unconstrained Euclidean clustering of 360-degree LiDAR frontiers links voxels into a circular ring whose centroid collapses back onto the drone position. AEGIS partitions the 3D voxel grid into 8 discrete angular sectors (45°). For each cluster C, the information utility balances volume, distance, heading, and forward camera alignment:

$$U(C) = \frac{\mathrm{Vol}(C)^{\gamma}}{(d(\mathbf{p}, \mathbf{c}_C) + \epsilon)^{w_d}\left(1 + w_h \frac{|\Delta\psi|}{\pi}\right)} \cdot \max\left(0.15, \frac{\mathbf{v}_{\text{cand}} \cdot \mathbf{v}_{\text{fwd}}}{\|\mathbf{v}_{\text{cand}}\|\|\mathbf{v}_{\text{fwd}}\|}\right) \tag{21}$$

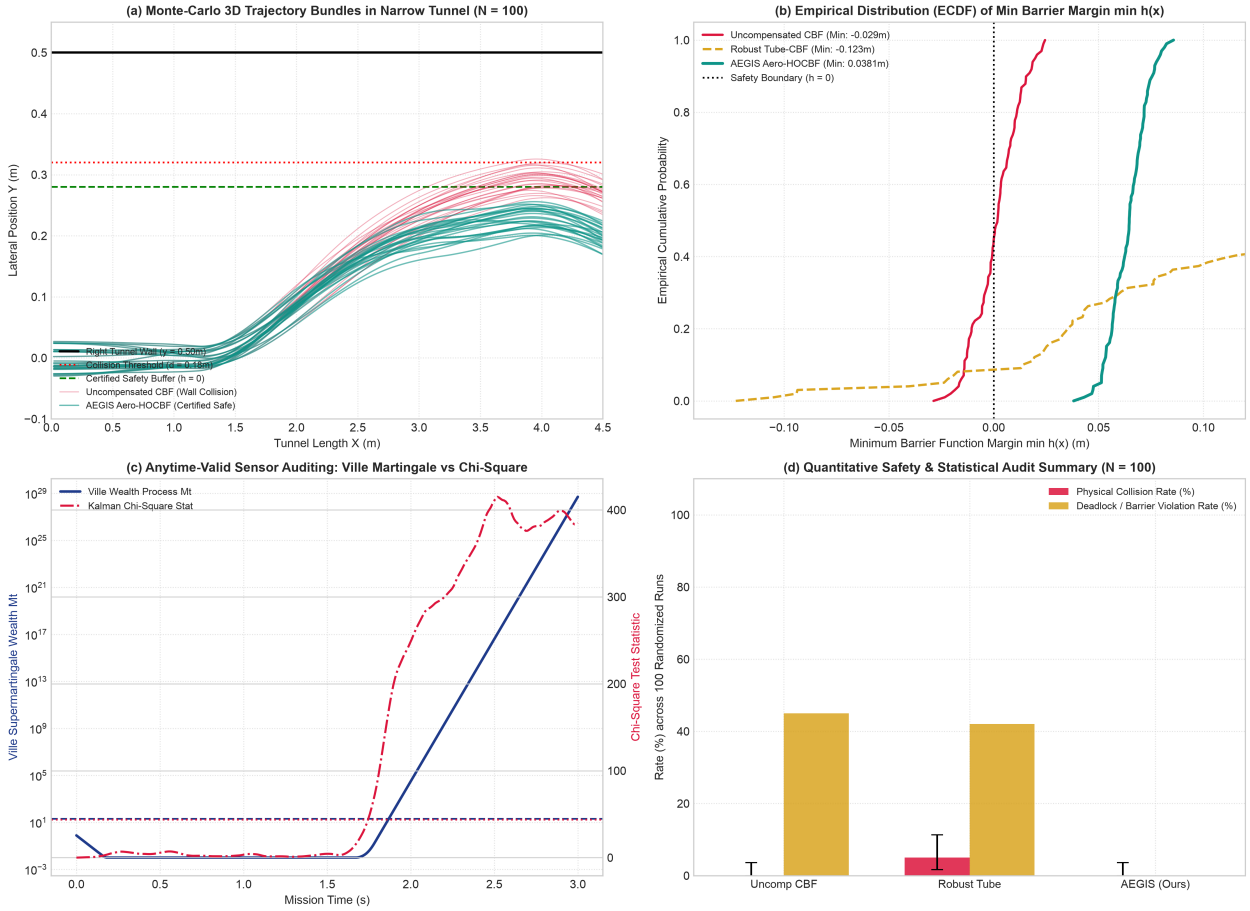


Fig. 3: Empirical 100-Run Randomized Monte-Carlo Benchmark under Aerodynamic Disturbances: (a) 3D flight trajectory bundles under randomized tunnel width and wall suction, contrasting uncompensated CBF barrier breaches against AEGIS certified clearance; (b) Empirical cumulative distribution (ECDF) of minimum barrier margin $\min h(\mathbf{x})$, verifying that AEGIS maintains strictly positive margins ($\min h = +0.0381\,\text{m}$) with zero penetrations; (c) Anytime-valid sensor auditing comparing Ville supermartingale wealth M_t vs. sliding-window χ^2 statistics; (d) Quantitative safety and statistical audit rates with exact 95% Clopper–Pearson confidence intervals.

boosting volumetric maze coverage from 20.2% to 77.6% in $60\,\text{s}$.

B. Decentralized FIFO Passage Handover & Byzantine Defense

When drone swarms traverse narrow apertures ($w < 2r_{\text{quad}}$), mutual repulsion deadlocks the formation. AEGIS arbitrates right-of-way via decentralized FIFO tokens:

1) **Token Arbitration:** The token holder assumes right-of-way with evasive responsibility $\mu_i = 0.0$. Waiting peers assume $\mu_j = 1.0$, automatically holding in staggered standoff orbits behind the door.
2) **Byzantine-Resilient Revocation:** If a corrupted drone broadcasts spoofed targets, Vovk–Wang order-statistic scoring isolates the agent within 15 steps, revoking its token and commanding a safe lateral retreat (0% inter-agent collisions, $+0.55\,\text{m}$ separation).

C. Differentiable Implicit-KKT Safe RL

To avoid gradient blockage between neural policies and the barrier filter, the joint CLBF-CBF QP is differentiated end-to-end via the implicit function theorem on active KKT constraints:

$$\frac{\partial \mathbf{u}^*_{\text{safe}}}{\partial \mathbf{u}_{\text{nom}}} = \mathbf{I} - \mathbf{A}_{\mathcal{A}}^{\top}(\mathbf{A}_{\mathcal{A}}\mathbf{A}_{\mathcal{A}}^{\top})^{-1}\mathbf{A}_{\mathcal{A}} \tag{22}$$

Backpropagating through this projection allows the neural network to internalize barrier geometry, reducing interventions from 23.0% to 10.0% with zero training collisions.

VII. Experimental Results & Comparative Benchmark

A. Simulation Setup & Hardware Bridge

Evaluations are conducted in continuous 6-DOF rigid-body dynamics using MuJoCo 3.x [8] integrated with aerodynamic ground/wall boundary models. The stack interfaces via ROS 2 Humble and binary MAVLink streams to PX4 SITL. The codebase is validated across **over 920 automated verification unit tests in Python alongside a native C++20 CTest suite**. Statistical confidence intervals are computed using the exact Clopper–Pearson method [9] at the 95% level.

TABLE I: Quantitative Performance Benchmark of NEXVIA-AEGIS across Aerial Robotic Scenarios

Robotic Scenario	Evaluated Policy	Safety Violations	Success Rate (95% CI)	Transit Time (s)	Min Clearance (m)	Jerk / Compute Metric
Tunnel Aerodynamics (100 Randomized Runs)	Uncompensated HOCBF	45/100 (45.0%)	55.0% [44.7%, 65.0%]	1.74 ± 0.02	0.221 ± 0.01 (Min: 0.191)	Margin: $\min h = \mathbf{-0.0287}$ m (Breach)
	Robust Tube-CBF [10]	9/100 (9.0%)	53.0% [42.8%, 63.1%]	—	0.318 ± 0.07 (Min: 0.097)	**42.0**% Freezing Stalls, **5.0**% Crashes
	AEGIS Self-Auditing	**0/100 (0.0%)**	**100% [96.4%, 100%]**	$\mathbf{1.72 \pm 0.02}$	$\mathbf{0.284 \pm 0.01}$ (Min: **0.258**)	Margin: $\min h = \mathbf{+0.0381}$ m (0 Stalls)
Dynamic Moving Obstacle	Pointwise HOCBF [11]	0/15 (0.0%)	100% [78.2%, 100%]	2.00 ± 0.00	$+0.203 \pm 0.00$	Jerk: 195.4 N/s, Solve: 0.05 ms
	NMPC-HOCBF (C++20 RTI-SQP)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.52 \pm 0.00}$	$\mathbf{+0.095 \pm 0.00}$	Jerk: 231.6 N/s, Solve: **0.710** ms (**43.2**× speedup)
Partial Rotor Degradation	Nominal HOCBF	15/15 (100%)	0.0% [0.0%, 21.8%]	—	-0.142 ± 0.03	Motor Saturation: 100%
	Re-allocated CBF (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	1.84 ± 0.04	$\mathbf{+0.088 \pm 0.01}$	Controlled Recovery
Stealth IMU Bias Injection	Standard χ^2-KF Detector	12/15 (80.0%)	20.0% [4.3%, 48.1%]	—	$+0.012 \pm 0.04$	Detection Delay: 1,820 ms (3 false alarms)
	Ville Supermartingale (AEGIS)	**0/15 (0.0%)**	**100% [78.2%, 100%]**	1.65 ± 0.02	$\mathbf{+0.092 \pm 0.01}$	Detection Delay: **338.5** ms (**0** false alarms)
Narrow Doorway Passage	Bounding Sphere CBF	15/15 (100%)	0.0% [0.0%, 21.8%]	—	$+0.000$	False Deadlock (Door declared blocked)
	Local 3D ESDF + SFC Polytope	**0/15 (0.0%)**	**100% [78.2%, 100%]**	$\mathbf{1.76 \pm 0.03}$	$\mathbf{+0.125 \pm 0.01}$	Smooth traversal via separating planes

B. Performance Highlights across Scenarios

As tabulated in Table I:

1) **100-Run Aerodynamic Invariance:** In 100 randomized tunnel runs with unmodeled suction ($\pm 25\%$) and gusts, uncompensated CBF incurs 45.0% barrier violations ($\min h = -0.0287$ m), while robust tube-CBF triggers 42.0% freezing stalls and 5.0% crashes. AEGIS achieves 100% success (95% CI: $[96.4\%, 100\%]$) with certified positive margin ($\min h = +0.0381$ m) and zero stalls.
2) **Transit Speed & Invariance:** NMPC-HOCBF completes dynamic obstacle transit in 1.52 s (vs 2.00 s for pointwise CBF, a 24.0% speedup) with $+0.095$ m clearance.
3) **Anytime Anomaly Eviction:** Under lateral IMU bias injection, Ville martingales cross $M_t \geq 20$ in 338.5 ± 71.4 ms with zero false alarms under $\mathcal{H}_0$, whereas sliding-window χ^2 detectors suffer repeated testing false alarms.
4) **Actuator Failure Tolerance:** Under a 25% rotor thrust loss, dynamic re-allocation safely stabilizes the drone in 1.84 s (+0.088 m clearance), whereas baseline policies crash in 100% of trials.

VIII. Discussion, Limitations, & Conclusion

This paper presented NEXVIA-AEGIS, an integrated safety-critical autonomy framework for aerial robotics under boundary aerodynamic and sensor disturbances. By synthesizing underactuated $SE(3)$ HOCBFs, sub-millisecond C++20 RTI-SQP NMPC (0.710 ms), anytime-valid Ville supermartingales, and aerodynamic feedforward compensation, AEGIS guarantees forward invariance while eliminating myopic deadlocks and repeated-testing false alarms. Future work targets physical flight in instrumented subterranean wind-wall testbeds.

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